

Linear Algebra and its Applications

Assignment 6

Due: 18 April 2024

In this assignment you will explore the application of (SVD) to an image classification problem using the MNIST dataset. It consists of 70,000 grayscale images of handwritten digits (0–9), each of size 28×28 . Each image can be viewed as a vector in $\mathbb{R}^{784}$ by unwrapping the pixel values.

Instructions

- Use Python (NumPy, matplotlib, etc.).
- Submit a PDF of your Jupyter Notebook including the code.
- Avoid including unnecessary outputs (errors, trial runs, etc.).

Problems

1. Data Preparation

Load the MNIST dataset. For each digit $n \in \{0, \dots, 9\}$, construct a matrix

$$D(n) \in \mathbb{R}^{784 \times N}$$

whose columns are vectorized images of that digit (exclude labels).

You may choose N between 2500 and 3000.

2. SVD and Principal Components

For each $D(n)$, compute the singular value decomposition:

$$D(n) = U\Sigma V^T.$$

Extract the first k left singular vectors (principal components).

- Display one sample image from the dataset.
- Display the first 4 principal components (reshape as 28×28 images).

Question: What features do the leading singular vectors appear to capture?

3. Projection-based Classification

For a test image $z \in \mathbb{R}^{784}$, define:

$$\rho_n(z) = \|z - U_k U_k^T z\|.$$

Classify z as:

$$\hat{n} = \arg \min_n \rho_n(z).$$

Questions:

- (a) Explain why this is equivalent to a least squares problem.
- (b) Why is this formulation computationally efficient?

4. Effect of Rank k

Repeat classification for different values of k :

$$k = 2, 4, 8, 16, 32.$$

Plot classification accuracy versus k .

Question: Why does accuracy improve and then saturate?

5. Energy Capture

For each $D(n)$ compute:

$$E_k = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_i \sigma_i^2}.$$

Question: How does energy relate to classification accuracy?

6. Noise Robustness

Add Gaussian noise to test images and repeat classification.

Question: Why might smaller k perform better in noisy settings?

7. Results

Include:

- Confusion matrix
- Accuracy plots
- Sample reconstructions